

DAY 03 - NETWORK INTERFACES

Networking Assignment Report

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Assignment Code	DAY03-NETWORK-INTERFACES
Total Marks	20
Operating System	Ubuntu Linux

Note: The assignment specifies the Windows command `ipconfig /all`. This practical was performed on Ubuntu Linux, so equivalent Linux networking commands were used.

THEORY

1. What is a Network Interface Card?

A Network Interface Card (NIC) is a hardware component that allows a computer or other device to connect to a network. It provides the interface through which data is transmitted and received. Modern computers usually have built-in Ethernet or Wi-Fi network interfaces.

2. What is Ethernet?

Ethernet is a widely used networking technology for connecting devices over a local area network (LAN). It commonly uses copper cables or fiber-optic cables and follows IEEE 802.3 standards.

3. What is RJ-45?

RJ-45 is the common 8-pin connector used with Ethernet twisted-pair network cables. It is used to connect Ethernet cables to network devices such as computers, switches, and routers.

4. What does UTP stand for?

UTP stands for Unshielded Twisted Pair. It is a type of copper networking cable made of pairs of insulated wires twisted together to reduce electromagnetic interference.

5. Name three common Ethernet cable categories.

Three common Ethernet cable categories are Cat5e, Cat6, and Cat6a. Cat6 and Cat6a generally support higher performance and better interference handling than older Cat5e installations.

6. What is a straight-through cable?

A straight-through Ethernet cable has the same wiring standard at both ends, such as T568A-to-T568A or T568B-to-T568B. It is traditionally used to connect different types of devices, such as a computer to a switch.

7. What is a crossover cable?

A crossover cable uses different wiring standards at its two ends, causing the transmit and receive pairs to be crossed. It was traditionally used to connect similar devices directly, such as computer-to-computer or switch-to-switch.

8. Explain Auto MDI-X.

Auto MDI-X is a feature that automatically detects the transmit and receive wiring needed for an Ethernet connection and adjusts the interface accordingly. Because of this, many modern devices can work with either straight-through or crossover cables.

9. Difference between copper and fiber-optic networking?

Copper networking carries electrical signals through metal conductors, while fiber-optic networking carries data as pulses of light through optical fiber. Fiber generally supports longer distances, higher bandwidth, and strong resistance to electromagnetic interference, while copper is usually cheaper and easier to install for short distances.

10. What is multimode fiber?

Multimode fiber (MMF) is an optical fiber designed to carry multiple light modes or paths at the same time. It is commonly used for shorter-distance network connections, such as inside buildings and data centers.

11. What is single-mode fiber?

Single-mode fiber (SMF) is an optical fiber with a small core that carries essentially one mode of light. It is suitable for long-distance, high-bandwidth communication and is widely used in telecommunications and long network links.

12. Two advantages of fiber-optic networking.

Two major advantages are: (1) high bandwidth and fast data transmission, and (2) immunity to electromagnetic interference. Fiber can also support much longer distances than typical copper Ethernet without the same level of signal loss.

13. Why is understanding network interfaces important for cybersecurity?

Understanding network interfaces helps cybersecurity professionals identify how a system connects to networks, inspect IP and MAC information, monitor traffic, troubleshoot connectivity, and recognize suspicious or unauthorized network activity. It is an important foundation for network security and incident analysis.

PRACTICAL IDENTIFICATION

14. Identify your active network adapter.

The active network adapter identified from the Ubuntu ip addr and ip route output is wlp0s20f3.

15. Write your IPv4 address, default gateway, and network adapter name.

Information	Value
Network Adapter	wlp0s20f3
IPv4 Address	10.84.163.114
Default Gateway	10.84.163.135

Linux commands used for the practical:

ip addr
ip route
iwconfig
ping google.com
traceroute google.com

Windows-to-Ubuntu equivalents: ipconfig → ip addr/ifconfig; ipconfig /all → ip addr + ip route + related interface information; tracert → traceroute.

SCREENSHOT EVIDENCE

The assignment requests an ipconfig /all screenshot. Since this report was completed on Ubuntu Linux, the screenshots below show the Linux equivalents used to identify the network interface and routing information.

```
md-hamza-khan@md-hamza-khan-HP-ZBook:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s31f6: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000
    link/ether e8:d8:d1:ec:91:9e brd ff:ff:ff:ff:ff:ff
3: wlp0s20f3: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default qlen 1000
    link/ether c8:09:a8:e6:1e:f5 brd ff:ff:ff:ff:ff:ff
    inet 10.84.163.114/24 brd 10.84.163.255 scope global dynamic noprefixroute wlp0s20f3
        valid_lft 2491sec preferred_lft 2491sec
    inet6 2409:40e3:5056:7da1:eea1:1382:fef1:396d/64 scope global temporary dynamic
        valid_lft 7173sec preferred_lft 7173sec
    inet6 2409:40e3:5056:7da1:1e42:32e7:aa2d:2e12/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 7173sec preferred_lft 7173sec
    inet6 fe80::e142:eb01:8f64:4d7d/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
4: wlx00e02691d31b: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default qlen 1000
    link/ether 00:e0:26:91:d3:1b brd ff:ff:ff:ff:ff:ff
md-hamza-khan@md-hamza-khan-HP-ZBook:~$ hostname -I
```

Figure 1 – Ubuntu ip addr output (network adapter and IPv4 address)

```
md-hamza-khan@md-hamza-khan-HP-ZBook:~/networking$ ip route
default via 10.84.163.135 dev wlp0s20f3 proto dhcp src 10.84.163.114 metric 600
10.84.163.0/24 dev wlp0s20f3 proto kernel scope link src 10.84.163.114 metric 600
md-hamza-khan@md-hamza-khan-HP-ZBook:~/networking$
```

Figure 2 – Ubuntu ip route output (default gateway and adapter)

```
md-hamza-khan@md-hamza-khan-HP-ZBook:~$ hostname -I
10.84.163.114 2409:40e3:5056:7da1:eaa1:1382:fef1:396d 2409:40e3:5056:7da1:1e42:32e7:aa2d:2e12
md-hamza-khan@md-hamza-khan-HP-ZBook:~$ ip route
default via 10.84.163.135 dev wlp0s20f3 proto dhcp src 10.84.163.114 metric 600
10.84.163.0/24 dev wlp0s20f3 proto kernel scope link src 10.84.163.114 metric 600
```

Figure 3 – Network information

```
md-hamza-khan@md-hamza-khan-HP-ZBook:~$ resolvectl status
Global
  Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
  resolv.conf mode: stub

Link 2 (enp0s31f6)
  Current Scopes: none
  Protocols: -DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported

Link 3 (wlp0s20f3)
  Current Scopes: DNS
  Protocols: +DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
  Current DNS Server: 2409:40e3:5056:7da1::f7
  DNS Servers: 10.84.163.135 2409:40e3:5056:7da1::f7

Link 4 (wlx00e02691d31b)
  Current Scopes: none
  Protocols: -DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
```

Figure 4 – Network information


```
md-hamza-khan@md-hamza-khan-HP-ZBook:~$ ping google.com
PING google.com (2404:6800:4002:817::200e) 56 data bytes
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=1 ttl=115 time=36.0 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=2 ttl=115 time=55.4 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=3 ttl=115 time=57.5 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=4 ttl=115 time=51.5 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=5 ttl=115 time=38.4 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=6 ttl=115 time=70.4 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=7 ttl=115 time=65.8 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=8 ttl=115 time=78.2 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=9 ttl=115 time=62.2 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=10 ttl=115 time=52.2 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=11 ttl=115 time=57.4 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=12 ttl=115 time=48.2 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=13 ttl=115 time=53.2 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=14 ttl=115 time=52.1 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=15 ttl=115 time=55.8 ms
64 bytes from del11s12-in-x0e.1e100.net (2404:6800:4002:817::200e): icmp_seq=16 ttl=115 time=51.3 ms
^C
--- google.com ping statistics ---
16 packets transmitted, 16 received, 0% packet loss, time 15005ms
rtt min/avg/max/mdev = 35.955/55.365/78.217/10.264 ms
```

Figure 5 – Network connectivity/practical evidence

```

md-hamza-khan@md-hamza-khan-HP-ZBook:~$ traceroute google.com
traceroute to google.com (142.250.182.14), 30 hops max, 60 byte packets
 1  _gateway (10.84.163.135)  2.302 ms  3.387 ms  4.262 ms
 2  * * *
 3  255.0.0.2 (255.0.0.2)  51.457 ms  54.517 ms  53.131 ms
 4  255.0.0.3 (255.0.0.3)  53.089 ms  50.498 ms  52.928 ms
 5  192.168.227.98 (192.168.227.98)  48.761 ms  192.168.227.99 (192.168.227.99)  49.272 ms  192.168.227.98 (192.168.227.98)  49.272 ms
 6  172.25.86.124 (172.25.86.124)  52.768 ms  172.25.119.228 (172.25.119.228)  31.816 ms  172.25.86.126 (172.25.86.126)  31.816 ms
 7  * * *
 8  * * *
 9  72.14.195.34 (72.14.195.34)  49.340 ms * *
10  192.178.83.245 (192.178.83.245)  48.374 ms  48.813 ms  31.347 ms
11  209.85.252.65 (209.85.252.65)  42.114 ms  41.808 ms  209.85.252.71 (209.85.252.71)  41.326 ms
12  tzdelb-bf-in-f14.1e100.net (142.250.182.14)  40.113 ms  209.85.252.71 (209.85.252.71)  39.360 ms  209.85.252.65 (209.85.252.65)  39.360 ms
md-hamza-khan@md-hamza-khan-HP-ZBook:~$

```

Figure 6 – Network practical evidence